Ex: 8
06/03/2024

TRANSIENT ANALYSIS OF SERIES RLC CIRCUIT

Aim:

To determine transient response of a series RLC circuit using Multisim Simulator

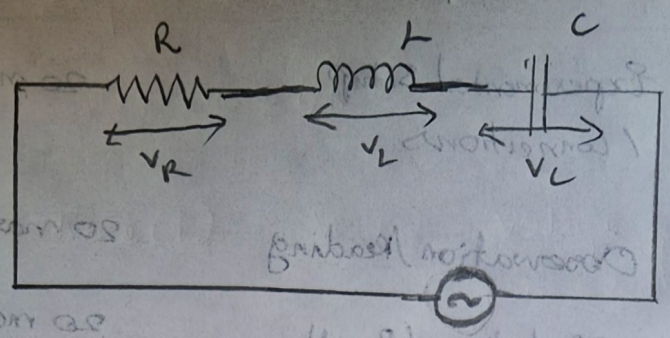
APPARATUS REQUIRED:

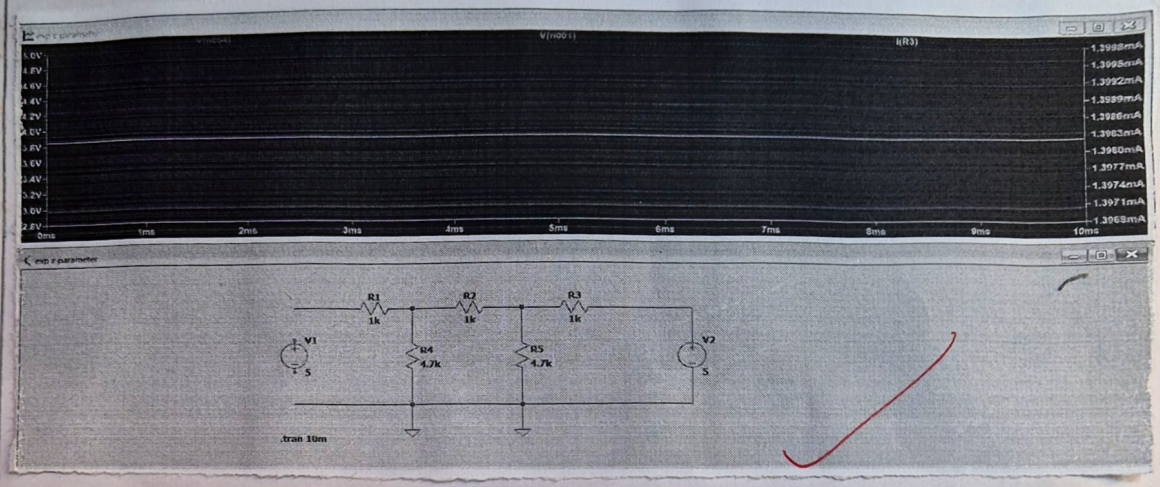
Sno	Components	Specification	Quantity
1.	CRO	10MHz	1
2.	DCB, BIB	---	1
3.	Resistors	10kΩ	
4.	Bread board	---	1
5.	Connecting wires	---	—
6	PC with Multisim Simulator	---	--

THEORY:

Series RLC circuit consist of a resistance, a capacitance and an inductance connected in series across an alternating supply. In a series RLC circuit containing a resistor, a inductor and a capacitor the source voltage V_{in} is the phasor sum made up of three components, V_R V_L and V_C with the current common to all three components it is used as the horizontal reference when constructing a voltage triangle.

CIRCUIT DIAGRAM:





Voltage across Resistor :

Time(t)	$V_R(t)$ - Voltage across resistor Observed value
$t = 0$	1.46 V
$t = \tau = 1ms$	6.34 V
$t = 2\tau = 2ms$	8.66 V
$t = 3\tau = 3ms$	9.44 V
$t = 4\tau = 4ms$	9.81 V
$t = 5\tau = 5ms$	9.855 V

The impedance of the circuit is the total opposition to the flow of current. For a series RLC circuit, and impedance triangle can be drawn by dividing each side of the voltage triangle by its current, I, The voltage drop across the resistive element is equal to I*R, the voltage is equal to I*Z The angle between Vs and I will be the phase angle, θ.

PROCEDURE:

1. Make connection as per the circuit diagram.
2. Calculate the values ~~and~~ of $V_C(t)$, $V_R(t)$ and write in table.

Compare the calculated value and measured values of voltage in both the cases.

MARK SPLIT-UP

Sl. No	Description	Mark Allotted	Mark obtained
1.	Experimental Setup / Connections	20 marks	18
2.	Observation / Reading	20 marks	18
3.	Calculation / Result	20 marks	18
4.	Viva	20 marks	20
5.	Record		92

Result:

Thus the transient analysis of series RLL circuit is done practically using Multisim Simulator.